

CR30 / ELECTRONIC CAM FIELD AUDIT

KOSME TOP II AD TANDEM 20/1056 CE SA4 E2 | Registration L05358 | Manual K966-749

Large-print field edition - black, white and grayscale only. No color carries diagnostic meaning.

STOP BEFORE TOUCHING: PHOTOGRAPH AND RECORD EVERY CARD, CONNECTOR AND DIP BANK. DO NOT ALTER DIP SWITCHES OR CLONE / COPY CARDS UNTIL THE ORIGINAL CONFIGURATION IS CAPTURED.

1. JOB AND MACHINE IDENTIFICATION

Audit date / time		Technician(s)	
Work order / event ID		Production line / location	
Product / bottle format		Current recipe / cam number	
Machine state on arrival	☐ Running ☐ Stopped ☐ Intermittent	Production impact	☐ Down ☐ Reduced ☐ Quality issue
Control/display identity	CR30 (page heading) / KET39 (manual index alias)	Platform/head count	20 documented; observed: _____

2. SAFETY AND EVIDENCE GATE

☐ Exact error photographed BEFORE reset	☐ CR30 screen and full panel photographed
☐ Every card / connector / DIP bank photographed BEFORE change	☐ Plant LOTO applied before guards / covers removed
☐ OEM minimum 1-minute wait observed after power isolation	☐ Required PPE / energized-work controls applied for justified live measurement
☐ Guards and safety switches restored before powered test	☐ One-change-at-a-time rule accepted

DIP NOTATION RULE / ORIGINAL CONFIGURATION NOTES: Record physical ON arrow/mark, bank orientation, switch count and exact left-to-right pattern. Do not assume binary convention until board marking or schematic confirms it.

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3. INITIAL FAULT SNAPSHOT - COMPLETE BEFORE RESETTING

Exact CR30 / HMI message	
Error number(s)	☐ 24 Software ☐ 25 Hardware ☐ 26 High Temp ☐ 27 COM ☐ 28 Faulty Line Other: _____
When fault occurs	☐ Power-up ☐ RUN before zero ☐ After zero ☐ During production ☐ After format change ☐ After clone
Scope	☐ One board/card ☐ Several boards ☐ All boards ☐ Unknown
Board / head identified	
Beacon / machine state	☐ Red-stop ☐ Yellow-warning ☐ Green-running ☐ Blue-reset ☐ RUN ☐ READY
Can error be reset?	☐ Yes - stays clear ☐ Yes - returns ☐ No Time to return: _____
Observed symptom	☐ No label ☐ Misplaced label ☐ Plate mismatch ☐ No centering ☐ No communication Other: _____

4. REPLACEMENT / CLONE HISTORY

Replacement card installed?	☐ No ☐ Yes	Date / technician	
Replacement board / head		DIP / address set to	
Source card used for clone		Source different from target?	☐ Yes ☐ No ☐ Unknown
Machine code entered		OEM candidate	05358 - CONFIRM ON DISPLAY
Head quantity entered		Expected candidate	20 - VERIFY BEFORE REUSE
Copy counter reached 0	☐ Yes ☐ No ☐ Unknown	Power interrupted?	☐ No ☐ Yes ☐ Unknown
Startup result	☐ RUN seen ☐ Machine zero ☐ READY seen	Failure point	

Exact sequence already performed and what happened

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5A. ELECTRONIC CAM CARD INVENTORY - CARDS / HEADS 1-5

Record BEFORE changing DIP switches, connectors or programs. Use the exact switch orientation seen on the board.

Head	Photo ID	DIP pattern AS FOUND	Decoded address	Power V	Comm?	Ve firmware	Tp temp C	Pc change?	En / Sx notes
1					Y / N			Y / N	
2					Y / N			Y / N	
3					Y / N			Y / N	
4					Y / N			Y / N	
5					Y / N			Y / N	

Address conflicts / firmware mismatches / abnormal temperatures / damaged connectors / card observations

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5B. ELECTRONIC CAM CARD INVENTORY - CARDS / HEADS 6-10

Record BEFORE changing DIP switches, connectors or programs. Use the exact switch orientation seen on the board.

Head	Photo ID	DIP pattern AS FOUND	Decoded address	Power V	Comm?	Ve firmware	Tp temp C	Pc change?	En / Sx notes
6					Y / N			Y / N	
7					Y / N			Y / N	
8					Y / N			Y / N	
9					Y / N			Y / N	
10					Y / N			Y / N	

Address conflicts / firmware mismatches / abnormal temperatures / damaged connectors / card observations

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5C. ELECTRONIC CAM CARD INVENTORY - CARDS / HEADS 11-15

Record BEFORE changing DIP switches, connectors or programs. Use the exact switch orientation seen on the board.

Head	Photo ID	DIP pattern AS FOUND	Decoded address	Power V	Comm?	Ve firmware	Tp temp C	Pc change?	En / Sx notes
11					Y / N			Y / N	
12					Y / N			Y / N	
13					Y / N			Y / N	
14					Y / N			Y / N	
15					Y / N			Y / N	

Address conflicts / firmware mismatches / abnormal temperatures / damaged connectors / card observations

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5D. ELECTRONIC CAM CARD INVENTORY - CARDS / HEADS 16-20

Record BEFORE changing DIP switches, connectors or programs. Use the exact switch orientation seen on the board.

Head	Photo ID	DIP pattern AS FOUND	Decoded address	Power V	Comm?	Ve firmware	Tp temp C	Pc change?	En / Sx notes
16					Y / N			Y / N	
17					Y / N			Y / N	
18					Y / N			Y / N	
19					Y / N			Y / N	
20					Y / N			Y / N	

Address conflicts / firmware mismatches / abnormal temperatures / damaged connectors / card observations

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6. COMMON POWER, SAFETY AND COMMUNICATION AUDIT

CHECK	AS FOUND	RESULT / MEASUREMENT	PHOTO / EVIDENCE	CHANGED?
Board-supply circuit breaker condition	OK / FAULT / N/A			NO / YES
CR30 supply: terminal 1 +24 V / terminal 2 common	OK / FAULT / N/A			NO / YES
Board power at first, middle and last card	OK / FAULT / N/A			NO / YES
Head safety switches fully closed	OK / FAULT / N/A			NO / YES
Plates coupled / applicable centering photo checks	OK / FAULT / N/A			NO / YES
CR30-to-head connectors fully seated	OK / FAULT / N/A			NO / YES
Sliding contacts clean, aligned and making contact	OK / FAULT / N/A			NO / YES
Flat signal cable free of tears / folds / pin damage	OK / FAULT / N/A			NO / YES
Flat-cable connectors seated and strain relieved	OK / FAULT / N/A			NO / YES
Power connectors seated; cable screws tight	OK / FAULT / N/A			NO / YES
Shared common / negative continuity verified	OK / FAULT / N/A			NO / YES
Duplicate DIP / network addresses ruled out	OK / FAULT / N/A			NO / YES
Card firmware versions compared / compatible	OK / FAULT / N/A			NO / YES
Metal particles / contamination inside boards checked	OK / FAULT / N/A			NO / YES
Board-to-motor cable insulation checked (Err 025)	OK / FAULT / N/A			NO / YES
Ventilation / card temperature checked (Err 026)	OK / FAULT / N/A			NO / YES

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7. ERROR 27 - ONE BOARD vs ALL BOARDS DIAGNOSTIC BRANCH

ONE BOARD REPORTS COM ERROR	ALL BOARDS REPORT COM ERROR
1. Verify power at that board. 2. Inspect power cable and connector. 3. Tighten applicable cable screws. 4. Inspect that board flat-cable connection. 5. Confirm UNIQUE DIP address. 6. Compare firmware and card data. 7. Replace only after these checks remain good.	1. Verify common board power and breaker. 2. Verify head safety switches. 3. Inspect CR30-head connections and sliding contacts. 4. Inspect flat cable. 5. Connect ONE board only and test. 6. Add boards individually until failure returns. 7. Investigate the board / cable segment that triggers failure. 8. Replace blocking board only after isolation proves it. 9. If still unresolved, OEM directs CR23/I and CR30 replacement.

Board-by-board isolation log - begin with one known / likely good board. Record the physical cable segment changed at each step.

Step	Board added / removed	Address	Power V	CR30 comm	Error returned?	Cable / connector segment	Time / notes
1				OK / FAIL	Y / N		
2				OK / FAIL	Y / N		
3				OK / FAIL	Y / N		
4				OK / FAIL	Y / N		
5				OK / FAIL	Y / N		
6				OK / FAIL	Y / N		

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7A. ERROR 27 - BOARD-BY-BOARD ISOLATION LOG (CONTINUED)

Step	Board added / removed	Address	Power V	CR30 comm	Error returned?	Cable / connector segment	Time / notes
7				OK / FAIL	Y / N		
8				OK / FAIL	Y / N		
9				OK / FAIL	Y / N		
10				OK / FAIL	Y / N		
11				OK / FAIL	Y / N		
12				OK / FAIL	Y / N		

Isolation conclusion - FIRST step that caused communication failure

Suspect board / cable / connector / power segment and evidence

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NETWORK-WIDE COPY CAN OVERWRITE EVERY ONLINE CARD. DO NOT COPY UNTIL SOURCE CARD, TARGET ADDRESS, POWER AND COMMUNICATION CHAIN ARE PROVEN.

8. CONTROLLED REPLACEMENT AND PROGRAM-COPY VERIFICATION

☐ Machine isolated; replacement installed; connectors verified	☐ Replacement card DIP address documented and UNIQUE
☐ Source card selected is known-good and is NOT replacement	Source address: _____ Target address: _____
Language selected: _____	Head quantity entered: _____ (candidate 20 - verify)
Machine code entered: _____ (candidate 05358 - verify)	☐ Copy started without communication error
☐ Decrementing copy counter reached ZERO	☐ NO power interruption / connector disturbance during copy
☐ CR30 reset and displayed RUN	☐ First machine zero was achieved
☐ CR30 displayed READY	☐ Card firmware and parameter readout compared after copy
☐ Production test completed at CONTROLLED speed	Test speed / duration: _____

Copy result / exact message / failed step

Post-copy functional test - products run, heads affected, label result

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9. CR27 SYNCHRONIZER AND MACHINE-ZERO AUDIT

CR27 supply	+12 V observed: _____ Common observed: _____	Display active?	YES / NO
Operating display	Fo active size: _____	Next plate advances?	YES / NO
th head / platform count	_____	Expected candidate	20 - VERIFY ACTUAL
Pt pulses / revolution	_____	Fo format	_____
Zero sensor sequence	[] Last plate start-photo [] Zero sensor [] Plate 1 start-photo	Plate 1 correctly established?	YES / NO
Loss-of-zero symptom	Not present / Suspected / Confirmed	Average-delay fallback suspected?	YES / NO
Inputs / output	T3 encoder: _____ T4 start photo: _____ T5 zero: _____ T8 group start: _____	Shared common verified?	YES / NO

CR27 functional topology: Start photocell -> CR27 -> adhesive-group start input. Zero sensor establishes plate 1. If zero is lost, CR27 may continue starts while using the average of programmed plate delays.

Machine-zero observations / exact symptom / evidence IDs

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9A. CR27 PLATE-SPECIFIC Pi RECORD - PLATES 1-10

Record ORIGINAL Pi before changing. OEM wording describes increasing delay as starting the label in advance.

Plate	Pi AS FOUND	Label position result	Changed to	Reason / outcome / photo ID
1		GOOD / EARLY / LATE		
2		GOOD / EARLY / LATE		
3		GOOD / EARLY / LATE		
4		GOOD / EARLY / LATE		
5		GOOD / EARLY / LATE		
6		GOOD / EARLY / LATE		
7		GOOD / EARLY / LATE		
8		GOOD / EARLY / LATE		
9		GOOD / EARLY / LATE		
10		GOOD / EARLY / LATE		

Pi observations / change justification / rollback value if needed

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9B. CR27 PLATE-SPECIFIC Pi RECORD - PLATES 11-20

Record ORIGINAL Pi before changing. OEM wording describes increasing delay as starting the label in advance.

Plate	Pi AS FOUND	Label position result	Changed to	Reason / outcome / photo ID
11		GOOD / EARLY / LATE		
12		GOOD / EARLY / LATE		
13		GOOD / EARLY / LATE		
14		GOOD / EARLY / LATE		
15		GOOD / EARLY / LATE		
16		GOOD / EARLY / LATE		
17		GOOD / EARLY / LATE		
18		GOOD / EARLY / LATE		
19		GOOD / EARLY / LATE		
20		GOOD / EARLY / LATE		

Pi observations / change justification / rollback value if needed

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10. CR30 CENTERING SETTINGS AUDIT

Cam number / name		R1 common lag	
Fu mode	MANUAL / AUTO	Me reference	CENTRE / EDGE
Mp multiplier	SL x1 / SL x4	Sp polarity	DARK / LIGHT
SL minimum spot length		Sr measured spot	
Sg threshold readout		Empty-plate test NET1	OFF / ON / UNKNOWN

CENTERING RULE: FINAL CENTERING POSITION = R1 + Cp. R1 IS COMMON TO ALL PLATFORMS; Cp IS PLATFORM-SPECIFIC. RECORD ORIGINAL VALUES BEFORE ADJUSTMENT.

Observed centering symptom / bottle format / cam selected / sensor condition

CR30 settings evidence / screen photo IDs / changes authorized

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10A. CR30 PLATFORM Cp RECORD - PLATFORMS 1-10

OEM Cp range: -11.2 to +11.2 degrees. Final centering position = R1 + Cp. Record ORIGINAL value before adjustment.

Platform	Cp AS FOUND	Observed centering	Changed to	Outcome / bottle-photo ID
1		GOOD / CW / CCW		
2		GOOD / CW / CCW		
3		GOOD / CW / CCW		
4		GOOD / CW / CCW		
5		GOOD / CW / CCW		
6		GOOD / CW / CCW		
7		GOOD / CW / CCW		
8		GOOD / CW / CCW		
9		GOOD / CW / CCW		
10		GOOD / CW / CCW		

Centering observations / photo IDs / change justification

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10B. CR30 PLATFORM Cp RECORD - PLATFORMS 11-20

OEM Cp range: -11.2 to +11.2 degrees. Final centering position = R1 + Cp. Record ORIGINAL value before adjustment.

Platform	Cp AS FOUND	Observed centering	Changed to	Outcome / bottle-photo ID
11		GOOD / CW / CCW		
12		GOOD / CW / CCW		
13		GOOD / CW / CCW		
14		GOOD / CW / CCW		
15		GOOD / CW / CCW		
16		GOOD / CW / CCW		
17		GOOD / CW / CCW		
18		GOOD / CW / CCW		
19		GOOD / CW / CCW		
20		GOOD / CW / CCW		

Centering observations / photo IDs / change justification

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11. FINDINGS, CORRECTIVE ACTION AND PRODUCTION VERIFICATION

Evidence-backed failure statement - WHAT failed, WHEN, and on WHICH board(s) / head(s)

Primary cause category	POWER / BREAKER-SAFETY / ADDRESS-DIP / CONNECTOR / FLAT CABLE / SLIDING CONTACT / BOARD / FIRMWARE-PROGRAM / ZERO-SYNC / CENTERING
Failed component / location	
Evidence proving cause	
Corrective action	
Parts used / part numbers	
Values changed	Original: _____ New: _____
Production verification	Product / format: _____ Speed: _____ Duration / count: _____ Result: _____
Recurrence check	SAME SHIFT / 24 HOURS / 7 DAYS / 30 DAYS Owner: _____

Unresolved findings / escalation required

Asset Graph event ID		Work order / service ticket	
Photos attached		OEM pages used	65-80
Technician signature		Date / time closed	
Reviewed by		Review date	

CLOSE-OUT RULE: A REPAIR IS NOT COMPLETE UNTIL PRODUCTION VERIFICATION IS RECORDED AND THE ORIGINAL / FINAL DIP, Pi, Cp AND PROGRAM VALUES ARE TRACEABLE.